PSO vs PSO - Stat Tests

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1 Imports

[1]:
```
import numpy as np
import pandas as pd
from scipy.stats import kruskal
import scikit_posthocs as sp
```

2 Logit, Firm 1

2.1 Data Loading

[2]:
```
model = 'logit' # row number = 0

avg_prices = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

deltas = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

rpdis = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

for i in range(1, 8):
    # load the dataframes
    df_shock_none = pd.read_csv(f"../results/run_{i}/shock_none/pso_vs_pso.csv")
```

```
    df_shock_a = pd.read_csv(f"../results/run_{i}/shock_a/pso_vs_pso_schemeA.
↪csv")
    df_shock_b = pd.read_csv(f"../results/run_{i}/shock_b/pso_vs_pso_schemeB.
↪csv")
    df_shock_c = pd.read_csv(f"../results/run_{i}/shock_c/pso_vs_pso_schemeC.
↪csv")

    # average prices
    avg_prices["shock_none"].append(df_shock_none.iat[0, 1])
    avg_prices["shock_a"].append(df_shock_a.iat[0, 1])
    avg_prices["shock_b"].append(df_shock_b.iat[0, 1])
    avg_prices["shock_c"].append(df_shock_c.iat[0, 1])

    # deltas
    deltas["shock_none"].append(df_shock_none.iat[0, 6])
    deltas["shock_a"].append(df_shock_a.iat[0, 6])
    deltas["shock_b"].append(df_shock_b.iat[0, 6])
    deltas["shock_c"].append(df_shock_c.iat[0, 6])

    # rpdis
    rpdis["shock_none"].append(df_shock_none.iat[0, 8])
    rpdis["shock_a"].append(df_shock_a.iat[0, 8])
    rpdis["shock_b"].append(df_shock_b.iat[0, 8])
    rpdis["shock_c"].append(df_shock_c.iat[0, 8])
```

2.2 Kruskal-Wallis Test - Average Prices

[3]:
```
h_stat, p_val = kruskal(
    avg_prices["shock_none"],
    avg_prices["shock_a"],
    avg_prices["shock_b"],
    avg_prices["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Average Prices")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on Average Prices
H-statistic: 2.08, P-value: 0.55692373, epsilon_squared: 0.0769
```

As the P-value is not significant (> 0.05), there is no significant difference between the average prices.

2.3 Post-Hoc Testing - Average Prices

If the Kruskal-Wallis P-value is not significant, the Dunn matrix is not worth considering.

2.4 Kruskal-Wallis Test - Deltas

[4]:
```
h_stat, p_val = kruskal(
    deltas["shock_none"],
    deltas["shock_a"],
    deltas["shock_b"],
    deltas["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Deltas")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
 ↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on Deltas
H-statistic: 25.36, P-value: 0.00001299, epsilon_squared: 0.9392
```

2.5 Post-Hoc Testing - Deltas

[5]:
```
dunn_matrix = sp.posthoc_dunn(
    [
        deltas["shock_none"],
        deltas["shock_a"],
        deltas["shock_b"],
        deltas["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)
```

```
          1         2         3         4
1  1.000000  0.007241  0.333856  0.000011
2  0.007241  1.000000  0.333856  0.333856
3  0.333856  0.333856  1.000000  0.007241
4  0.000011  0.333856  0.007241  1.000000
```

- `deltas["shock_none"]` is significantly different from `deltas["shock_a"]`
- `deltas["shock_none"]` is significantly different from `deltas["shock_c"]`
- `deltas["shock_b"]` is significantly different from `deltas["shock_c"]`

2.6 Kruskal-Wallis Test - RPDIs

[6]:
```
h_stat, p_val = kruskal(
    rpdis["shock_none"],
    rpdis["shock_a"],
    rpdis["shock_b"],
    rpdis["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on RPDIs")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
 ↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on RPDIs
H-statistic: 25.46, P-value: 0.00001239, epsilon_squared: 0.9428
```

2.7 Post-Hoc Testing - RPDIs

[7]:
```
dunn_matrix = sp.posthoc_dunn(
    [
        rpdis["shock_none"],
        rpdis["shock_a"],
        rpdis["shock_b"],
        rpdis["shock_c"]
    ], p_adjust='holm'
)

print(dunn_matrix)
```

```
          1         2         3         4
1  1.000000  0.007089  0.331799  0.000010
2  0.007089  1.000000  0.331799  0.331799
3  0.331799  0.331799  1.000000  0.007089
4  0.000010  0.331799  0.007089  1.000000
```

- `rpdis["shock_none"]` is significantly different from `rpdis["shock_a"]`
- `rpdis["shock_none"]` is significantly different from `rpdis["shock_c"]`
- `rpdis["shock_b"]` is significantly different from `rpdis["shock_c"]`

3 Hotelling, Firm 1

3.1 Data Loading

[8]:
```
model = 'hotelling' # row number = 1

avg_prices = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

deltas = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

rpdis = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

for i in range(1, 8):
    # load the dataframes
    df_shock_none = pd.read_csv(f"../results/run_{i}/shock_none/pso_vs_pso.csv")
    df_shock_a = pd.read_csv(f"../results/run_{i}/shock_a/pso_vs_pso_schemeA.
 ↪csv")
    df_shock_b = pd.read_csv(f"../results/run_{i}/shock_b/pso_vs_pso_schemeB.
 ↪csv")
    df_shock_c = pd.read_csv(f"../results/run_{i}/shock_c/pso_vs_pso_schemeC.
 ↪csv")

    # average prices
    avg_prices["shock_none"].append(df_shock_none.iat[1, 1])
    avg_prices["shock_a"].append(df_shock_a.iat[1, 1])
    avg_prices["shock_b"].append(df_shock_b.iat[1, 1])
    avg_prices["shock_c"].append(df_shock_c.iat[1, 1])

    # deltas
    deltas["shock_none"].append(df_shock_none.iat[1, 6])
    deltas["shock_a"].append(df_shock_a.iat[1, 6])
    deltas["shock_b"].append(df_shock_b.iat[1, 6])
```

```
    deltas["shock_c"].append(df_shock_c.iat[1, 6])

    # rpdis
    rpdis["shock_none"].append(df_shock_none.iat[1, 8])
    rpdis["shock_a"].append(df_shock_a.iat[1, 8])
    rpdis["shock_b"].append(df_shock_b.iat[1, 8])
    rpdis["shock_c"].append(df_shock_c.iat[1, 8])
```

3.2 Kruskal-Wallis Test - Average Prices

[9]:
```
h_stat, p_val = kruskal(
    avg_prices["shock_none"],
    avg_prices["shock_a"],
    avg_prices["shock_b"],
    avg_prices["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Average Prices")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
 ↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on Average Prices
H-statistic: 3.00, P-value: 0.39162518, epsilon_squared: 0.1111
```

As the P-value is not significant (> 0.05), there is no significant difference between the average prices.

3.3 Post-Hoc Testing - Average Prices

If the Kruskal-Wallis P-value is not significant, the Dunn matrix is not worth considering.

3.4 Kruskal-Wallis Test - Deltas

[10]:
```
h_stat, p_val = kruskal(
    deltas["shock_none"],
    deltas["shock_a"],
    deltas["shock_b"],
    deltas["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Deltas")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
 ↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on Deltas
H-statistic: 0.66, P-value: 0.88229143, epsilon_squared: 0.0245
```

As the P-value is not significant (> 0.05), there is no significant difference between the deltas.

3.5 Post-Hoc Testing - Deltas

If the Kruskal-Wallis P-value is not significant, the Dunn matrix is not worth considering.

3.6 Kruskal-Wallis Test - RPDIs

[11]:
```
h_stat, p_val = kruskal(
    rpdis["shock_none"],
    rpdis["shock_a"],
    rpdis["shock_b"],
    rpdis["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on RPDIs")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on RPDIs
H-statistic: 3.54, P-value: 0.31579421, epsilon_squared: 0.1311
```

As the P-value is not significant (> 0.05), there is no significant difference between the RPDIs.

3.7 Post-Hoc Testing - RPDIs

If the Kruskal-Wallis P-value is not significant, the Dunn matrix is not worth considering.

4 Linear, Firm 1

4.1 Data Loading

[13]:
```
model = 'linear' # row number = 2

avg_prices = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

deltas = {
```

```
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

rpdis = {
    "shock_none": [],
    "shock_a": [],
    "shock_b": [],
    "shock_c": []
}

for i in range(1, 8):
    # load the dataframes
    df_shock_none = pd.read_csv(f"../results/run_{i}/shock_none/pso_vs_pso.csv")
    df_shock_a = pd.read_csv(f"../results/run_{i}/shock_a/pso_vs_pso_schemeA.
 ↪csv")
    df_shock_b = pd.read_csv(f"../results/run_{i}/shock_b/pso_vs_pso_schemeB.
 ↪csv")
    df_shock_c = pd.read_csv(f"../results/run_{i}/shock_c/pso_vs_pso_schemeC.
 ↪csv")

    # average prices
    avg_prices["shock_none"].append(df_shock_none.iat[2, 1])
    avg_prices["shock_a"].append(df_shock_a.iat[2, 1])
    avg_prices["shock_b"].append(df_shock_b.iat[2, 1])
    avg_prices["shock_c"].append(df_shock_c.iat[2, 1])

    # deltas
    deltas["shock_none"].append(df_shock_none.iat[2, 6])
    deltas["shock_a"].append(df_shock_a.iat[2, 6])
    deltas["shock_b"].append(df_shock_b.iat[2, 6])
    deltas["shock_c"].append(df_shock_c.iat[2, 6])

    # rpdis
    rpdis["shock_none"].append(df_shock_none.iat[2, 8])
    rpdis["shock_a"].append(df_shock_a.iat[2, 8])
    rpdis["shock_b"].append(df_shock_b.iat[2, 8])
    rpdis["shock_c"].append(df_shock_c.iat[2, 8])
```

4.2 Kruskal-Wallis Test - Average Prices

[14]:
```
h_stat, p_val = kruskal(
    avg_prices["shock_none"],
    avg_prices["shock_a"],
    avg_prices["shock_b"],
```

```
    avg_prices["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Average Prices")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
 ↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on Average Prices
H-statistic: 3.00, P-value: 0.39162518, epsilon_squared: 0.1111
```

As the P-value is not significant (> 0.05), there is no significant difference between the average prices.

4.3 Post-Hoc Testing - Average Prices

If the Kruskal-Wallis P-value is not significant, the Dunn matrix is not worth considering.

4.4 Kruskal-Wallis Test - Deltas

[16]:
```
h_stat, p_val = kruskal(
    deltas["shock_none"],
    deltas["shock_a"],
    deltas["shock_b"],
    deltas["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on Deltas")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
 ↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on Deltas
H-statistic: 23.56, P-value: 0.00003088, epsilon_squared: 0.8725
```

4.5 Post-Hoc Testing - Deltas

[17]:
```
dunn_matrix = sp.posthoc_dunn(
    [
        deltas["shock_none"],
        deltas["shock_a"],
        deltas["shock_b"],
        deltas["shock_c"]
    ], p_adjust='holm'
)
```

```
print(dunn_matrix)
```

```
          1         2         3         4
1  1.000000  0.032577  0.645388  0.000125
2  0.032577  1.000000  0.086480  0.214522
3  0.645388  0.086480  1.000000  0.000734
4  0.000125  0.214522  0.000734  1.000000
```

- `deltas["shock_none"]` is significantly different from `deltas["shock_a"]`
- `deltas["shock_none"]` is significantly different from `deltas["shock_c"]`
- `deltas["shock_b"]` is significantly different from `deltas["shock_c"]`

4.6 Kruskal-Wallis Test - RPDIs

[18]:
```
h_stat, p_val = kruskal(
    rpdis["shock_none"],
    rpdis["shock_a"],
    rpdis["shock_b"],
    rpdis["shock_c"]
)
n = 4 * 7 # total number of observations across all groups
epsilon_squared = h_stat / (n - 1)

print("Kruskal-Wallis Test on RPDIs")
print(f"H-statistic: {h_stat:.2f}, P-value: {p_val:.8f}, epsilon_squared:␣
 ↪{epsilon_squared:.4f}\n")
```

```
Kruskal-Wallis Test on RPDIs
H-statistic: 6.06, P-value: 0.10888130, epsilon_squared: 0.2243
```

As the P-value is not significant (> 0.05), there is no significant difference between the RPDIs.

4.7 Post-Hoc Testing - RPDIs

If the Kruskal-Wallis P-value is not significant, the Dunn matrix is not worth considering.